Orthogonality and Gram–Schmidt: Answer Key

Problem sheet for the Day 9 class · CSIR NET / GATE previous year questions

Answers

Q	1	2	3	4	5	6	7	8	9	10
Answer	(D)	(C)	(A)	(A)	(B)	(A),(B),(C),(D)	(A),(C)	(C),(D)	(C),(D)	(B),(D)

Hints and short solutions

1. **(D)** GATE 2011 $v_1 = \frac{1}{\sqrt{2}}(1,1,0)$. $u_2 - \langle u_2, v_1\rangle v_1 = (\frac{1}{2}, -\frac{1}{2}, 0)$, normalised $\frac{1}{\sqrt{2}}(1,-1,0)$. $u_3 - \sqrt{2}\,v_1 - 0\cdot v_2 = (0,0,1)$.
2. **(C)** GATE 2011 For the standard inner product the matrix of T^* is the conjugate transpose M^* of $M = \begin{pmatrix} 1 & -i & 0 \\ i & 1 & 0 \\ 1 & 1 & i \end{pmatrix}$, i.e. $M^* = \begin{pmatrix} 1 & -i & 1 \\ i & 1 & 1 \\ 0 & 0 & -i \end{pmatrix}$.
3. **(A)** GATE 2013 (A) gives $T\big(\sum_{n\le N} e_n/n\big) = \big(\sum_{n\le N} \frac{1}{n}\big)e_1$, unbounded. (B), (D) are shifts; (C) is bounded diagonal.
4. **(A)** GATE 2012 P is the definition of T^*; Q is P with the roles swapped and conjugated. R forces $T = T^*$; S fails for $T = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$: $\|Te_2\| = 1$, $\|T^*e_2\| = 0$.
5. **(B)** GATE 2012 $\bar{M}^T = M$, so M is Hermitian with real eigenvalues, and $(iM)^* = -iM$: iM is skew-Hermitian with purely imaginary eigenvalues.
6. **(A),(B),(C),(D)** CSIR NET June 2014 Gram matrix $G = \begin{pmatrix} 4 & 0 & 1 \\ 0 & 4 & -1 \\ 1 & -1 & 4 \end{pmatrix}$. $\|w+v-u\|^2 = 4+4+4+2(-1)-2(1)-0 = 8$. $\langle w, 4u-w\rangle = 4-4 = 0$. $\det G = 56 \neq 0$ gives independence.
7. **(A),(C)** CSIR NET Dec 2013 (A) is $y^\perp$; (C) is $(y^\perp)^\perp$. (B) misses 0; (D) is the single vector $y/\|y\|^2$.
8. **(C),(D)** CSIR NET June 2015 A is the Householder reflection in $u^\perp$: $A^2 = I - 4uu' + 4u(u'u)u' = I$, so A is invertible and $A^2 = A$ fails. $\operatorname{tr} A = n - 2\,u'u = n-2$.
9. **(C),(D)** CSIR NET June 2018 $S^{\perp\perp}$ is a closed subspace, so (A),(B) fail for $S = \{e_1, 2e_1\}$. (D): apply $T \subseteq T^{\perp\perp}$ to $T = S$ and $T = S^\perp$. (C) holds in finite dimension, as the official key assumes.
10. **(B),(D)** CSIR NET June 2017 $(AA^t)^{r-1}\cdot AA^t = I$, so AA^t is invertible and $\operatorname{rank} A = m \le n$. $A = (1\ 0)$ gives $AA^t = 1$ with $m \neq n$ and A^tA singular. If $m = n$, $\det A \neq 0$.